American Family Physician

Volume 90, Number 5 • September 1, 2014



A Comprehensive Newborn Examination: Part I. General, Head and Neck, Cardiopulmonary

Part II. Skin, Trunk, Extremities, Neurologic

Table 1. Normal Vital Signs in Newborns Born at 40 Weeks' Gestation

Vital sign	*Normal range*
Heart rate	120 to 160 beats per minute*
Respiratory rate	40 to 60 breaths per minute
Systolic blood pressure	60 to 90 mm Hg†
Temperature	97.7°F to 99.5°F (36.5°C to 37.5°C)‡
Weight	Females: 3.5 kg (7 lb, 12 oz); range, 2.8 to 4.0 kg (6 lb, 3 oz to 8 lb, 14 oz)
	Males: 3.6 kg (8 lb); range, 2.9 to 4.2 kg (6 lb, 7 oz to 9 lb, 5 oz)
Length	20 in (51 cm); range, 19 to 21 in (48 to 53 cm)
Head circumference	14 in (35 cm); range, 13 to 15 in (33 to 37 cm)

神經學檢查

1. 觀察自然臥姿：姿態多樣而富變化，當頭偏向一側時，和臉同側之肢體外張，或兩上肢曲張，膝部張開，腳掌朝內（圖 53）。四肢活動自如，上肢較下肢活躍，靜態休息時握拳，時而張開。當肢體活動伸直時，手掌和腳掌外張。若姿態鮮少變化，需留意有無神經方面異常。



自然臥姿

一般外觀檢查

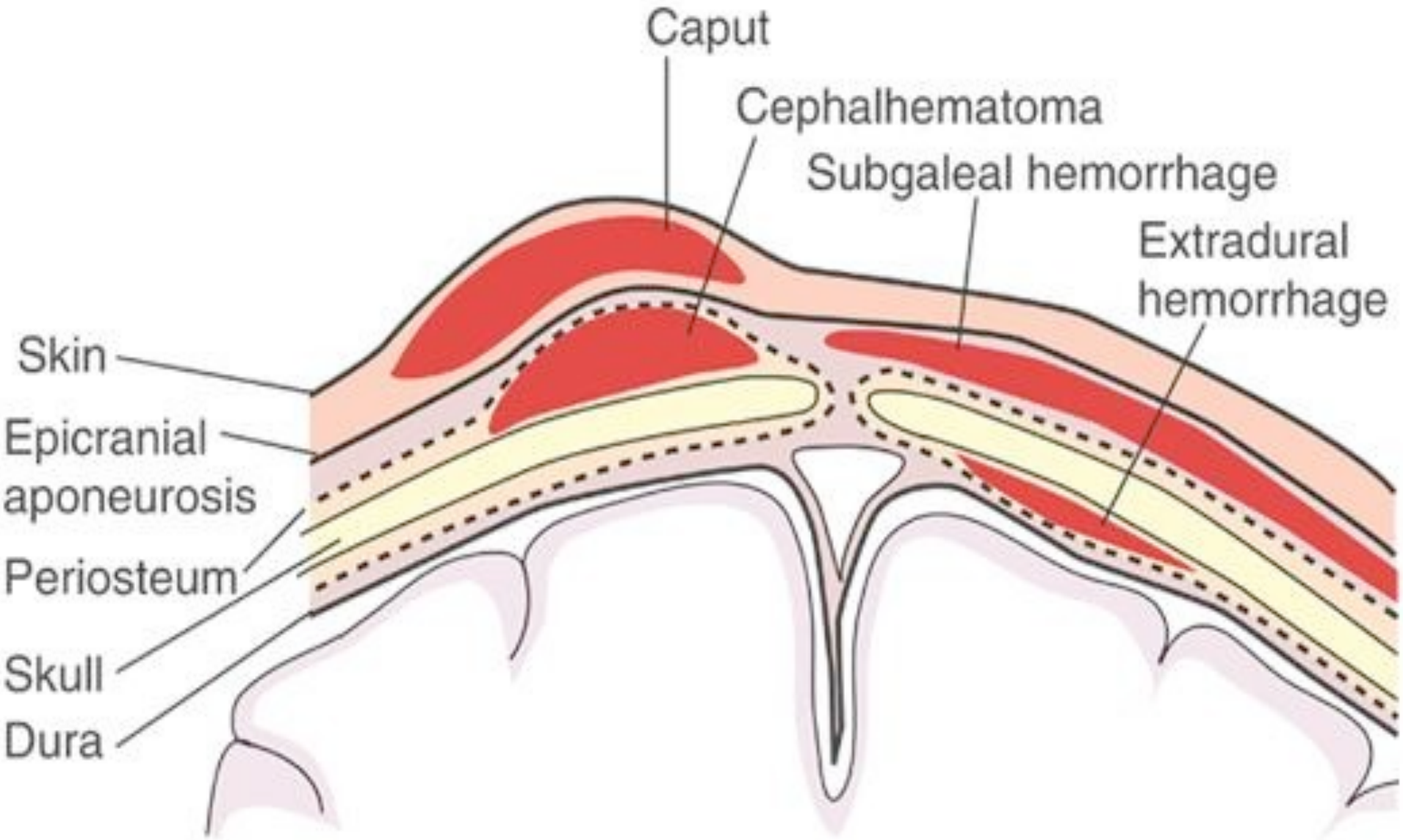
- **Skin color:**
  - Reddish pink – healthy warm babies
  - Generalized redness of the skin, plethora -- ploycythemia
  - Dry, cracked, or peeling skin – post-maturity
  - Vernix – whitish, most abundant in premature infants and is less prominent closer to term.
  - Greenish – meconium
  - Bruising and petechiae – birth process or need further evaluation
  - Pallor – anemia or peripheral shutdown with shock or both
  - Jaundice – normally, it appears between 2 and 4 days.
  - Mottled skin – cutis marmorata
  - Hemangioma, Mongolian spots
  - Erythema toxicum, milia

Table 1. Common Skin Findings in Newborns

Finding	*Characteristics*	*Location*
Salmon patch (nevus simplex)	Macular pink-red capillary dilations, often bilateral, fades over time	Forehead, upper eyelid, nape of the neck (stork bite)
Port-wine stain (nevus flammeus)	Dark purple or red, capillary malformation, generally does not fade, ophthalmology referral needed if near the eye	Nonspecific location Sturge-Weber syndrome: over the trigeminal nerve (associated with seizures, glaucoma) Klippel-Trénaunay-Weber syndrome: extremity
Hemangioma	Benign vascular tumor, caused by increased growth of endothelial cells on blood vessels Often benign and self-involutes, management depends on size and location	Nonspecific
Erythema toxicum	Flesh-colored papules with erythematous base, contains eosinophils, resolves in first week	Diffuse (face and trunk)
Pustular melanosis	Pustules without erythema, contain neutrophils, rupture and leave hyperpigmented macules that may persist for months	Diffuse (forehead, chin, neck, back)
Milia	White papules, consist of epidermal cysts with keratinous material, spontaneous rupture within first few weeks of life	Nose
Mongolian spots (dermal melanocytosis)	Blue-gray macules, up to 10-cm diameter, more common in black, Native American, and Hispanic populations; usually fade by four years of age, but can be confused with abuse and should be documented	Common in sacral region
Café au lait spots	Light-brown macule, likely benign but may be an early sign of neurofibromatosis or McCune-Albright syndrome	Nonspecific
Cutis marmorata	Reticular mottling of the skin caused by vascular response to cold, typically resolves with warmth	Usually localized to the lower limbs

一般外觀檢查

- **Head:**
  - Molded – normal
  - Anterior fontanel -- tension and size
  - Cranial sutures -- up to 1 cm is normal
  - Caput succedaneum -- edema caused by pressure over the presenting part; crosses suture lines
  - Cephalohematoma -- collections of blood between the periosteum and the skull bones; most in the parietal; do not extend across the suture lines
  - Subgaleal hemorrhage – fluctuant; blood accumulation beneath the scalp in a large space

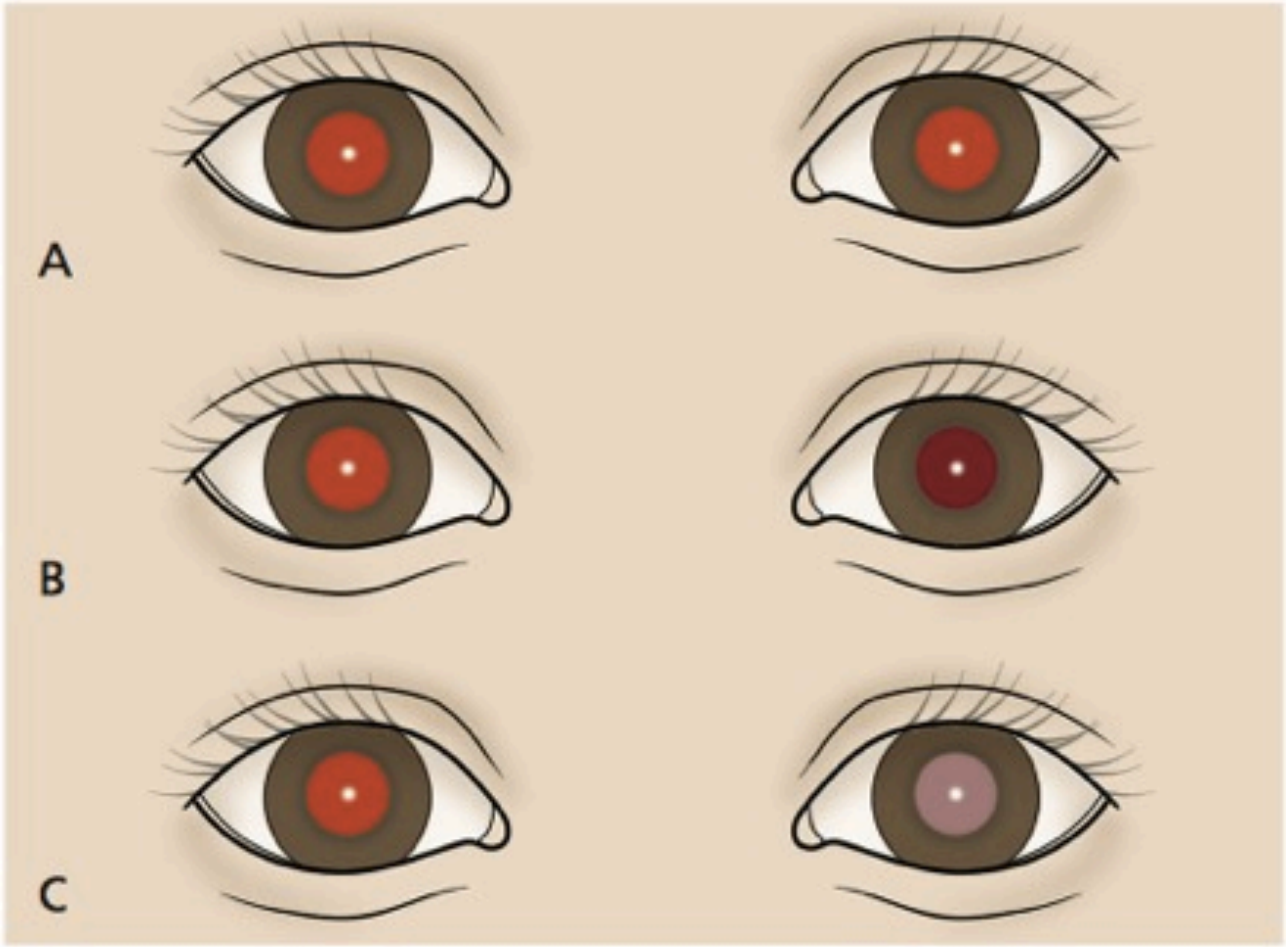


(Adapted from Pape KE, Wigglesworth JS: Haemorrhage, Ischemia and the Perinatal Brain. Philadelphia, JB Lippincott, 1979.)

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一般外觀檢查

- **Eyes:** Hold the infant upright and note
  - haziness or clouding of the cornea, lens.
  - Proptosis, squint or asymmetric closure (facial nerve palsy) of the lids
  - Red reflex (abnormal in retinoblastoma, congenital cataract)
  - Subconjunctival hemorrhages: very common after birth



ILLUSTRATIONS BY MARCIA HARTSOCK

Figure 3. Interpretation of red reflex test results in a newborn. *(A)* Normal, symmetric red reflex. *(B)* Normal red reflex in the right eye, and abnormal, diminished red reflex in the left eye, which is most commonly caused by refractive error between the eyes, but can also be caused by a more serious pathology (e.g., retinoblastoma). *(C)* Normal red reflex in the right eye and no reflex in the left eye, which occurs when the reflection is blocked by an opacity such as a cataract.

一般外觀檢查

- **Nose:**
  - One can assess the patency of the posterior choanae by holding the mouth closed and listening with the stethoscope for the outrush of each naris.

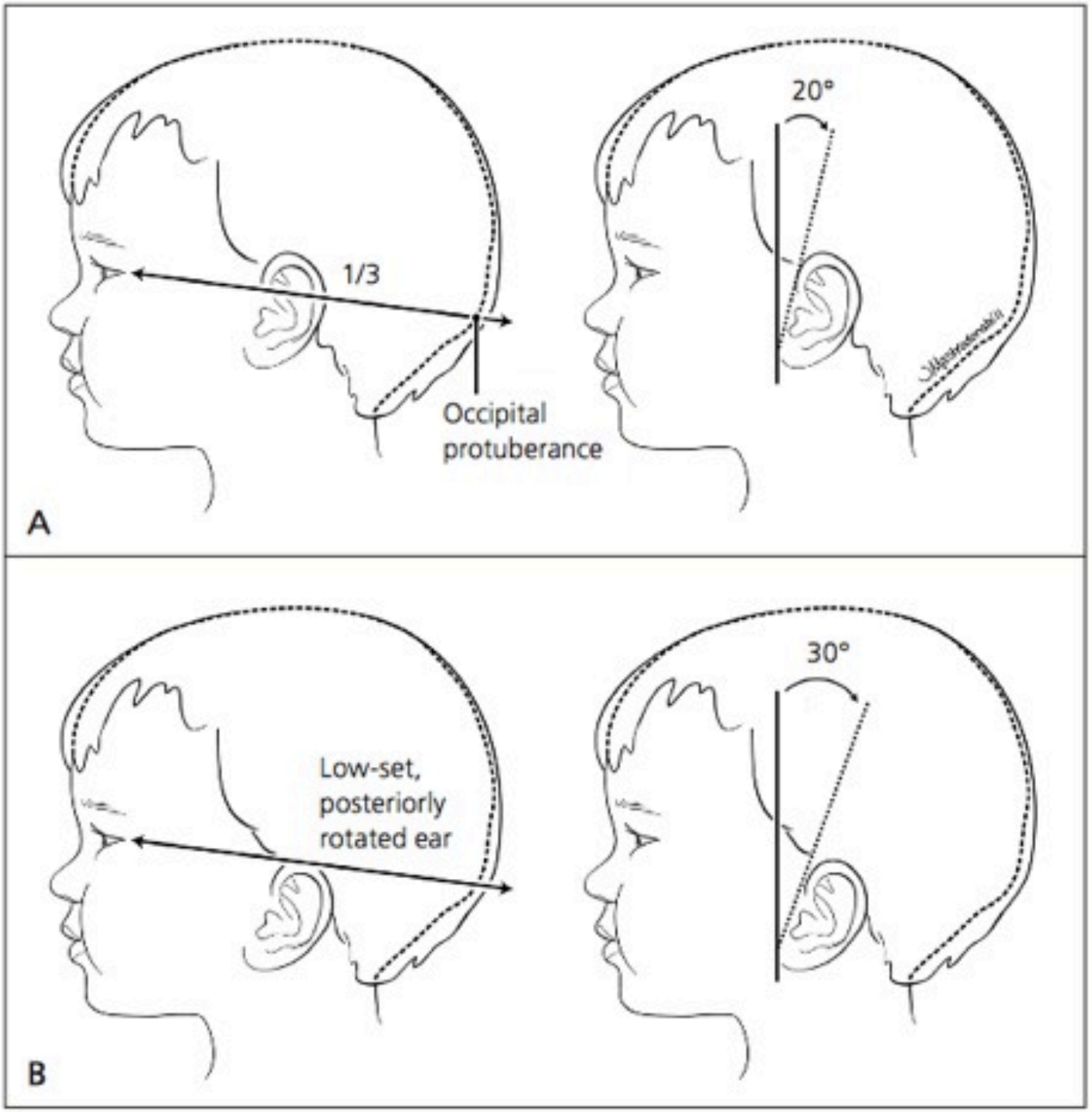


Figure 4. Evaluation of ear position in a newborn. *(A)* Normal ear position. *(B)* Low-set, posteriorly rotated ear, which may be a sign of a genetic condition, such as Down, Turner, or trisomy 18 syndrome.

Ears:

- The ears can be grossly malformed, uncommonly large or small, angled abnormally or set lower on the head than normal.
- Very low placement plus unusual size, floppiness and perpendicularity to the skull suggest renal agenesis or chromosomal aberration



Normally developed outer ear (pinna)



Abnormal size, shape, rotation and/or location of pinna

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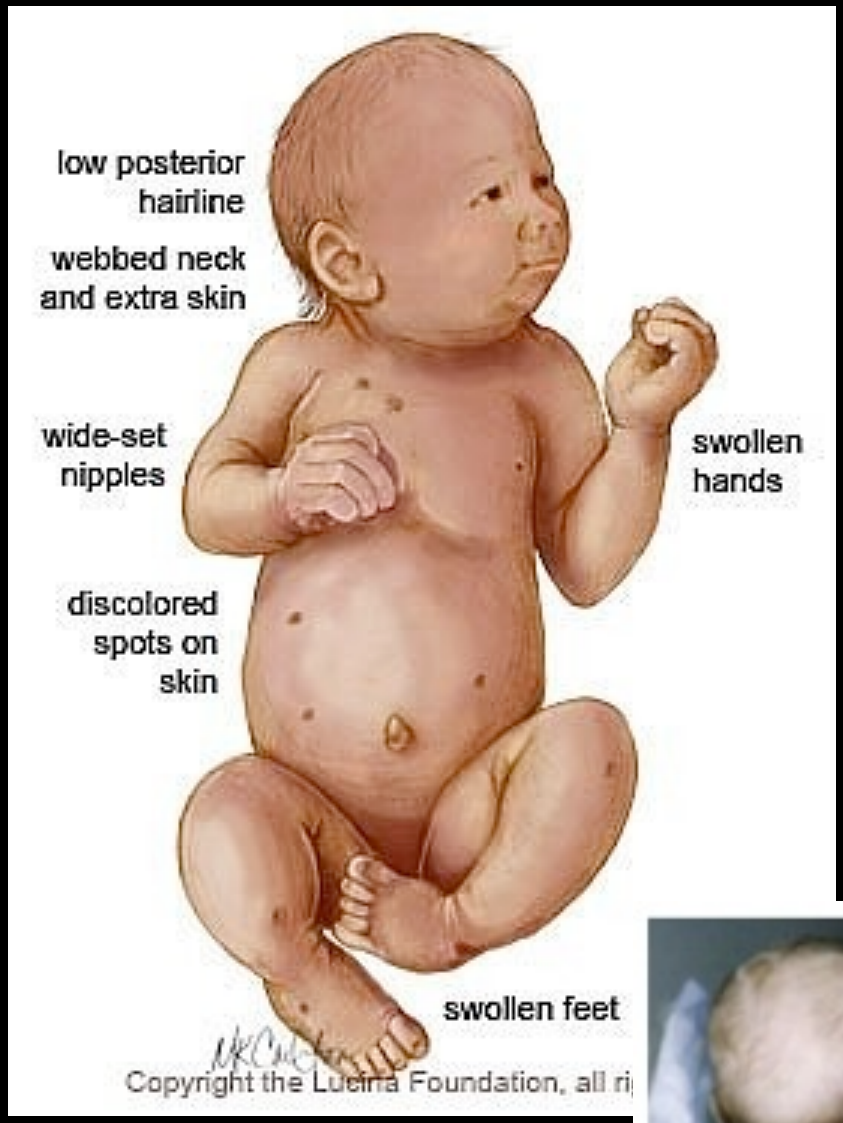
一般外觀檢查

- **Mouth:**
  - Note clefts of lip and palate (examine entire palate)
  - symmetric movement of lip corners
  - Look for retention cysts along the alveolar ridge
  - Note the size of the tongue and depress the lower jaw or take advantage of a cry to see the posterior pharynx.
  - Examine the frenulum and note its length.

一般外觀檢查

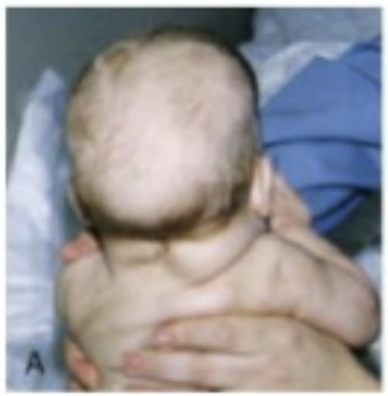
- **Neck:**
  - Note length and mobility and inspect for congenital cysts, hygromas, thyroglossal duct cysts, and thyromegaly
  - Look for webbing and palpate the length of both clavicles to rule out fracture.
  - Sternomastoid tumors: hematoma in the sternomastoid muscle

Turner syndrome **SGA**













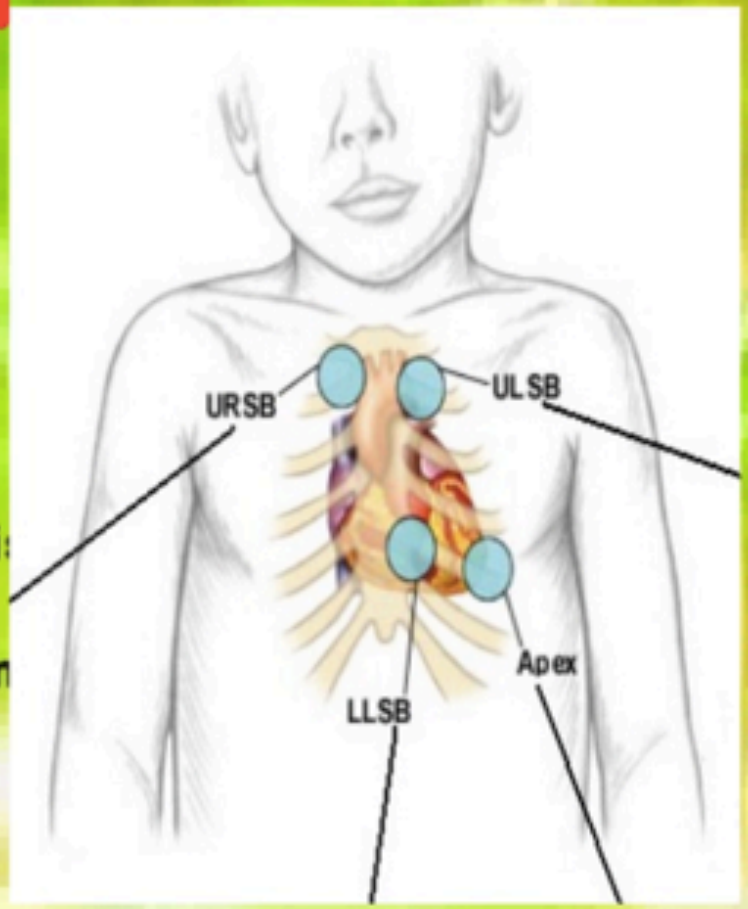
視診、觸診、聽診、扣診

- **Chest:**
  - Inspect for overinflation, symmetric movement, presence or absence of retractions and the use of accessory muscles.
  - Auscultate for rales, rhonchi, and bowel sounds.
  - Respiratory rate: normally 40-60 breaths/minute

視診、觸診、聽診、扣診

- **Cardiovascular system:**
  - Palpate the precordium: thrills or pronounced ventricular heave
  - Point of maximal impulse: usually found in the L't 4th intercostal space inside the mid-clavicular line
  - Heart rate: abnormally fast (>160/min) or slow (<100/min)
  - The presence or absence of murmurs
  - Palpability of the arm, femoral and distal arterial pulses

Location :



Systolic:

Aortic Valve Stenosis

Diastolic:

Aortic Regurgitation

Systolic:

1. Pulmonary valve stenosis(TOF)
2. Atrial septal defect
3. Pulmonary ejection murmur
4. Innocent Pulmonary flow murmur of newborn
5. Pulmonary artery stenosis
6. Aortic stenosis
7. Coarctation of the aorta
8. Patent ductus arteriosus
9. Partial & total anom. pulm. venous return .

Diastolic:

Pulmonary Regurgitation

Systolic:

1. Ventricular septal Defect including ECD
2. Vibratory innocent Murmur (Still's murmur)
3. HOCM
4. Tricuspid Regurgitation

Diastolic:

Tricuspid stenosis

Systolic:

1. Mitral Regurgitation
2. Vibratory Innocent Murmur(still's M)
3. Mitral valve Prolapse Syndrome
4. Aortic stenosis
5. HOCM

Diastolic:

視診、觸診、聽診、扣診

- **Abdomen:**
  - Look for unusual flatness (diaphragmatic hernia) or excessive fullness (one must then determine if this is due to an excess of air within or outside the bowels, to excess fluid, to an enlarged viscous or viscera or to the presence of a cystic or solid tumor).
  - Visible gastric or bowel patterns may be considered an almost certain sign of obstruction.
  - Umbilicus: three vessels
    - any discharge or reddening of the skin
    - Green discoloration: meconium stain => intrauterine asphyxia
    - Note umbilical hernia

視診、觸診

- **Genitalia** - Male
  - size and formation of the penis
  - position of the meatus
  - size of the scrotum and the nature of its skin
  - descent of non-descent of the testes
  - palpate over the inguinal canals for presence of hernia
  - Anus area

視診、觸診

- **Genitalia** – female
  - size of the clitoris
  - the nature of the skin of the labia majora
  - Vulva: the clitoris and labia minora are relatively prominent
  - the position of the vaginal and urethral orifices
  - palpate over the inguinal canals for presence of hernia
  - Anus area

Back and hips:

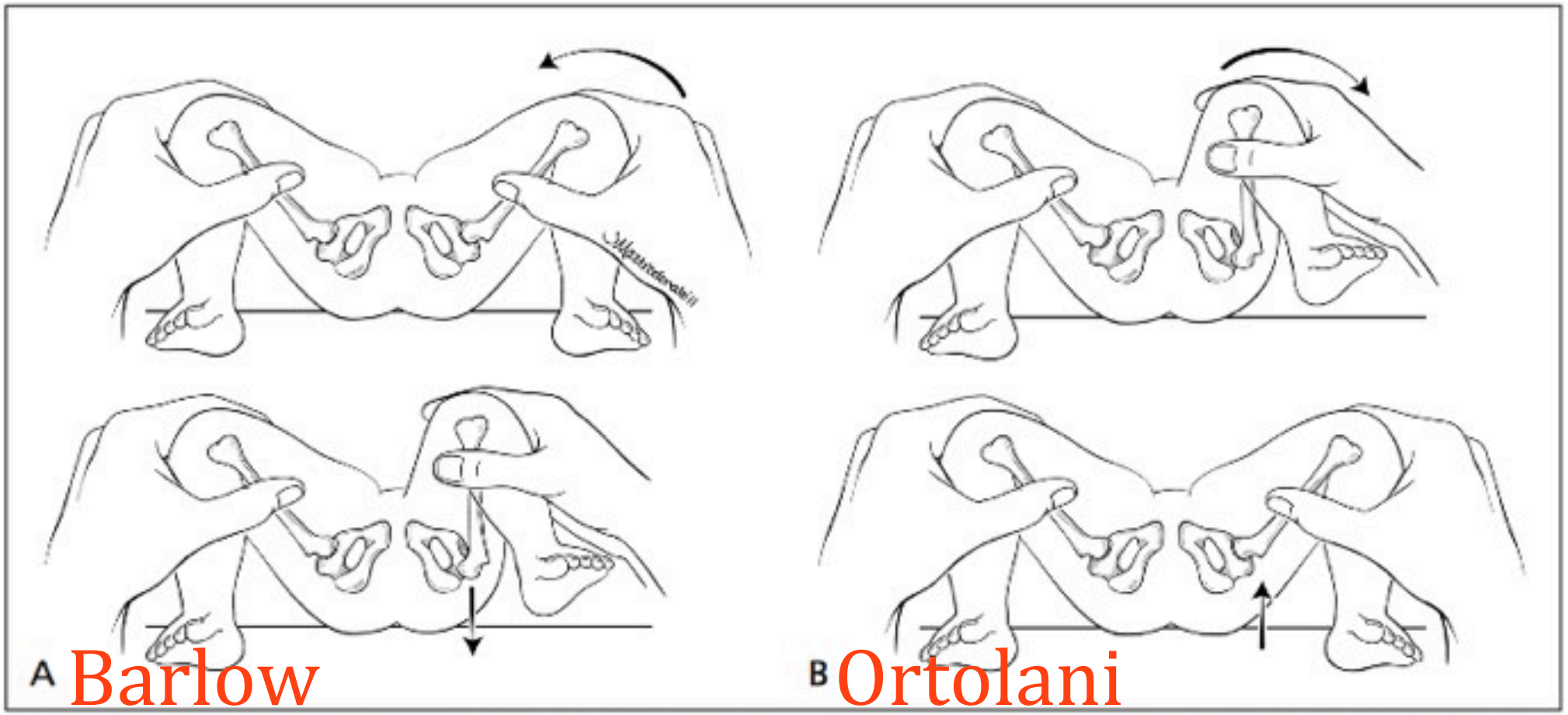


Figure 3. Techniques to evaluate for hip dysplasia in newborns. *(A)* The Barlow maneuver involves adducting the hip while pushing the thigh posteriorly to see if can be dislocated. *(B)* The Ortolani maneuver involves abducting the hips while pushing the thigh anteriorly and relocates reducible hips dislocated by the Barlow maneuver creating a clunk.

一般外觀檢查

- **Extremities:**
  - Shape, posture, symmetry
  - Syndactyly or polydactyly
  - Observe spontaneous movements
  - **Erb's palsy** -- lack of movement in the arm and the hand showed 'waiter's tip' position
  - 'Rocker bottom' shape and short hallux → Edward syndrome (Trisomy 18)
  - Calcaneovalgus position → fetal position in utero

神經學檢查

- Formal testing is seldom needed
- **Posture**: normal neonate
  -- lies predominantly in a flexed position
- **Spontaneous motor activity**
- **Muscle tone** and strength: test by
  - Assessing resistance to passive movements
  - Pull-to-sit maneuver
  - Ventral suspension
- **Crying**: note if it is either-high pitched or very weak
- **Feeding and sucking patterns**

Table 2-4. Timing of Primitive Reflexes

Reflex	Present	Disappears
Moro	At birth (term)	6 mo
Suck	At birth	6–8 mo
Root	At birth	6–8 mo
Place and step	At birth	1–2 mo
Tonic neck	Birth–2 mo	6–7 mo
Reflex grasp	At birth	2–3 mo
Lateral prop	4–5 mo	Does not disappear
Parachute	8–9 mo	Does not disappear

From Algranati PS: Pediatric examination. In Willms JL, Schneiderman H, Algranati PS (eds): Physical Diagnosis: Bedside Evaluation of Diagnosis and Function. Baltimore, Williams & Wilkins, 1994, p 665.

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3. 牽引反應（traction response）：將嬰兒自仰臥姿勢拉起至垂直的坐姿，可見頭部後仰（圖 55 左），但會有短暫向前挺直的動作。越過鉛垂線後頭部向前傾，軀幹為弧形（圖 55 右）。如果後仰太厲害，需小心追蹤是否發育遲緩。



圖 55 牽引反應

4. 腹部懸抱（ventral suspension）：檢查者以手支撐嬰兒腹部，滿一個月時，嬰兒的頭部和軀幹會形成一直線，髖部半張開，下肢自然半曲張下垂（圖56）。如下肢僵直，或軀幹過軟呈倒 U 字型，則為異常。



圖 56 腹部懸抱

5. 擁抱反射（Moro reflex）：如圖用手撐住頭部，突然讓頭部下降 2.5 公分，正常的嬰兒會有雙手對稱性向前外張的動作（圖 57）。重複測試反射會愈來愈不明顯，通常出生時即有，5~6 個月大消失。



圖 57 擁抱反射

10. 不對稱頸部張力反射（asymmetric tonic neck reflex）：將頭轉向一側，可見同側的手臂伸直，而另一手臂屈曲如騎馬射箭般的姿勢。此反射在 2 個月後變弱，5~6 個月大消失（圖 61）。



不對稱頸部張力反射

11. 跨步與抬步反射（stepping and placing reflex）：立姿抱住，下壓足部碰觸地面，會有向前跨步走的反射性動作，稱為跨步反射（圖 62）。以桌緣刺激足背，有登步的反射性動作，稱為抬步反射，這些反射在 1 個月後變弱。





SORT: KEY RECOMMENDATIONS FOR PRACTICE

Clinical recommendation	*Evidence rating*
Screening for hypoglycemia should be performed in newborns who are large or small for gestational age, newborns of mothers with diabetes mellitus, and late preterm infants (34 to 36 6/7 weeks gestational age).	C
Regardless of red reflex test results, all newborns with a family history of retinoblastoma, cataracts, glaucoma, or retinal abnormalities should be referred to an ophthalmologist who is experienced in the examination of children.	C
Hearing should be evaluated in all newborns before one month of age, but preferably before discharge, using the auditory brainstem response or the otoacoustic emissions test.	C
Recent data indicate that ultrasonography should be performed in newborns with isolated ear anomalies, such as preauricular pits or cup ears, only when they are associated with one or more of the following characteristics: other malformations or dysmorphic features, teratogenic exposures, a family history of deafness, or a maternal history of gestational diabetes.	C
Routine screening for congenital heart disease via pulse oximetry is recommended before discharge at 24 hours of life or later. Diagnostic echocardiography should be performed if screening results are positive.	C

SORT: KEY RECOMMENDATIONS FOR PRACTICE

Clinical recommendation	*Evidence rating*
Routine renal ultrasonography to look for renal anomalies in newborns with isolated cases of a single umbilical artery is not beneficial.	C
Bilateral undescended testes, a micropenis, or a bifid scrotum should prompt investigation for ambiguous genitalia.	C
Newborns with a hypospadias should not be circumcised because the foreskin may be needed for repair.	C
A sacral dimple is simple if it is less than 0.5 cm in diameter, located within 2.5 cm of the anal verge, and not associated with cutaneous markers (discoloration or hypertrichosis). In the absence of these criteria, ultrasonography should be performed before three months of age to evaluate for spinal dysraphism.	C
Girls born in the breech position should receive imaging to evaluate for hip dysplasia. Imaging should be considered in newborns with a family history of developmental hip dysplasia and in newborn boys born in the breech position.	C

A = consistent, good-quality patient-oriented evidence; B = inconsistent or limited-quality patient-oriented evidence; C = consensus, d evidence, usual practice, expert opinion, or case series. For information about the SORT evidence rating system, go to http://www.aafp.